

Chain-of-thought prompting elicits reasoning in large language models (Wei et al., 2022)

Read Abstract, Section 1 (Introduction), Section 2 (Chain-of-Thought Prompting), and Figure 1 (examples).

Reading Activities

Questions

1. What exactly is added to a prompt to make it a 'chain-of-thought' prompt?
Reproduce the structure from Figure 1 in your own words.

It's not entirely clear to me because the authors do not explain it well, but it looks like chain-of-thought prompting involves giving the LLM a model of how it should reason through a given task before asking it to engage in a similar task.

2. Wei et al. report that CoT benefits are 'emergent' — they appear only in sufficiently large models. What does this imply for practitioners using smaller or older LLMs?

This would suggest that if you're using a smaller or older LLM, CoT procedures may not work well or may not work at all.

3. The paper tests CoT on arithmetic, symbolic reasoning, and commonsense reasoning tasks. Which of these task types is most analogous to tasks you perform professionally?

I suppose commonsense reasoning would be the most similar to the work I do, though I don't think that's a good description and I generally don't find the term "commonsense" to be useful.

4. What is the key limitation of the few-shot CoT approach identified by the authors? Does this concern apply to your planned Task 2 use case?

It's outside of the sections we were asked to read, but at the end of Section 6, the authors stated that we can't know whether the LLM (they use the term "neural network" is actually reasoning or not.

Key Terms

Term	My Definition
Chain-of-thought prompting	providing a model that an LLM can use to guide its own “reasoning” in solving a problem
Zero-shot prompting	not defined in the text, but it appears to be a situation in which the LLM can successfully complete the task with minimal context/instructions provided
Few-shot prompting	not defined in the text, but it appears to be a task that an LLM can complete successfully with a few prompts to provide context, information, and or correction (as opposed to just one in zero-shot)
Reasoning chain	a set of steps that indicate (or mimic) problem solving when an LLM is working on a task/problem
Emergent ability	a capability which arises naturally in very large LLMs in which they become able to complete tasks that they were not specifically designed to accomplish

Survey of hallucination in natural language generation (Ji et al., 2023)

Read Abstract, Section 1, and Section 2. Skip Sections 3–7 on first pass.

Reading Notes

tolerance of hallucinations can be different in different tasks (e.g., in a summary, you want no hallucinations, but in a generated dialogue, you’re basically asking for creativity, which more or less consists of hallucinations)

Reading Activities

Questions

1. How does Ji et al. formally define hallucination in NLG? What is the relationship between 'faithfulness' and 'factuality' in their framework?

Their formal definition is “generated content that is nonsensical or unfaithful to the provided source content.” They described “faithfulness” as the consistency that an LLM response shows relative to the source material it is provided, while “factuality” is the objective provableness of its response or aspects of its response.

2. What is the difference between intrinsic and extrinsic hallucination? Give one example of each that could occur in an academic professional document.

An intrinsic hallucination is one which is inconsistent with the source material provided. In an example of summarizing student course/session feedback, this could take the form of reporting student feedback that students did not actually submit (i.e., was not in the source file).

An extrinsic hallucination is one which cannot be verified from the source material provided. In the same student feedback example as above, this could take the form of providing unsolicited recommendations for incorporating student feedback.

3. The paper identifies several causes of hallucination in the training process. Which cause do you think is hardest to mitigate through prompt engineering alone?

This question is addressed in Section 3, which was outside of our assigned reading, but I think that imperfect representation learning would be difficult to mitigate through prompt engineering since it's a process internal to the LLM. Erroneous decoding would also be a candidate, but since part of that (focusing on the wrong part of the input) could probably be dealt with through prompt engineering, I think imperfect representation would be the most problematic.

4. Based on the taxonomy in Section 2, what type of hallucination would you most need to guard against in the professional tasks you identified in your Module 1 reflection?

I think I would be more concerned about extrinsic hallucination, though I'd be worried about both. I'd like to think that I might be able to use an LLM to detect intrinsic hallucinations in a two-step process: first, with a well-crafted prompt, ask the LLM to generate the feedback summary; and second, in a different-well-crafted prompt, ask the LLM (or a different one?) to look for inconsistencies between the source data and the output from the first step. But that same process would probably also deal with extrinsic hallucinations. I guess a manual read-through might do a better job of picking up the more egregious intrinsic hallucinations, though.

Key Terms

Term	My Definition
Hallucinations (NLG)	incorrect or undesired information generated by an LLM in response to a prompt information generated by an LLM which is inconsistent with provided content and/or does not make sense (note that with this definition, there is such a thing as a factually correct hallucination)
Intrinsic hallucination	a hallucination which is inconsistent the source content provided
Extrinsic hallucination	a hallucination which is unrelated to the source content provided (and hence cannot be immediately verified and would require additional steps to verify), though Ji et al. note that the hallucination may not be incorrect
Faithfulness	the extent to which a generated response is consistent with the source material provided (not to be confused with factuality)
Factually grounding	assuming this is the same thing as factuality, this is the objective verifiability of LLM-generated output (or parts of output)